Dynamic Trend & Event Detector: Emerging Topic Detection & News Correlation using Neural Topic Modeling

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Course: Advanced Machine Learning and Deep Learning — Phase 1 & 2

Abstract

We address distinguishing fleeting social media trends from significant real-world events by tracking semantic velocity across longitudinal text corpora. Using 11,000 HuffPost articles (2012–2022), we implement a two-phase architecture. Phase 1 applies TF-IDF baseline extraction and LDA topic modeling with three novel temporal features, discovering COVID-19 and Trump/President topics unsupervised. Failure analysis traces topic conflation to LDA's bag-of-words assumption. Phase 2 resolves this via BERTopic — Sentence-BERT embeddings (384-dim), UMAP dimensionality reduction, HDBSCAN density clustering, and c-TF-IDF representation — discovering 15 topics with correct COVID-19 temporal placement (1,036% velocity in 2020), verified ACA/COVID context separation (cosine similarity 0.27 vs BoW ∼0.80), and confirmed via ablation study that contextual embeddings alone account for a 6-year temporal correction in COVID topic placement. We propose a Phase 3 hybrid approach to address remaining limitations.

1 Introduction

The exponential expansion of digital media renders it functionally impossible to manually distinguish superficial, fleeting social media trends from substantive, lasting real-world events. We introduce semantic velocity as our core mathematical concept — the rate of change of topic prevalence in a text corpus over time. While standard topic models provide thematic extraction, they fundamentally fail on continuous temporal streams because they treat all documents as temporally identical. As Allan et al. [1] established, event detection requires chronological contextualization. Our two-phase methodology bridges classical probabilistic models (LDA) with modern neural approaches (BERTopic), proving that contextual embeddings resolve semantic conflation that bag-of-words models cannot.

2 Literature Review

1. Allan et al. (1998): Topic Detection and Tracking pilot study [1] — established event detection parameters but relied on simple word overlap without temporal modeling.
2. Blei, Ng, Jordan (2003): LDA [2] revolutionized probabilistic topic modeling but is fundamentally static — ignores time via full exchangeability assumption.
3. McInnes et al. (2018): UMAP [5] preserves both local and global topological structure during dimensionality reduction via cross-entropy optimization on fuzzy simplicial sets — unlike t-SNE which only preserves local structure.
4. Reimers & Gurevych (2019): Sentence-BERT [7] introduced Siamese BERT networks. Standard BERT requires both sentences fed simultaneously — finding the most similar pair in n=10,000 sentences requires $n(n-1)/2 \approx$ 50 million computations (∼65 hours on V100). SBERT reduces this to 5 seconds by pre-computing fixed-size embeddings via mean pooling, while maintaining BERT-level accuracy on STS benchmarks.
5. Grootendorst (2022): BERTopic [4] combined SBERT + UMAP + HDBSCAN + c-TF-IDF. Its key innovation over Top2Vec is replacing centroid-based word selection with class-based TF-IDF — a cluster need not be spherical, so c-TF-IDF captures discriminative vocabulary regardless of cluster geometry.

Our work bridges the gap between LDA and BERTopic by explicitly engineering temporal features, providing rigorous comparative analysis, and proposing a hybrid Phase 3 architecture that addresses limitations of both phases.

3 Dataset and EDA

The HuffPost News Category Dataset (Misra [6]) contains 210,000+ articles across 42 categories spanning 2012–2022. We isolate a balanced subset of exactly 11,000 articles (1,000/year) to prevent temporal volume bias — election years have significantly more articles, which would artificially inflate velocity metrics if uncontrolled. Mean article length: 29 words. EDA reveals political coverage grew 278% during the 2016–2017 US election cycle — a non-obvious pattern that validated our entire temporal tracking approach before modeling began.

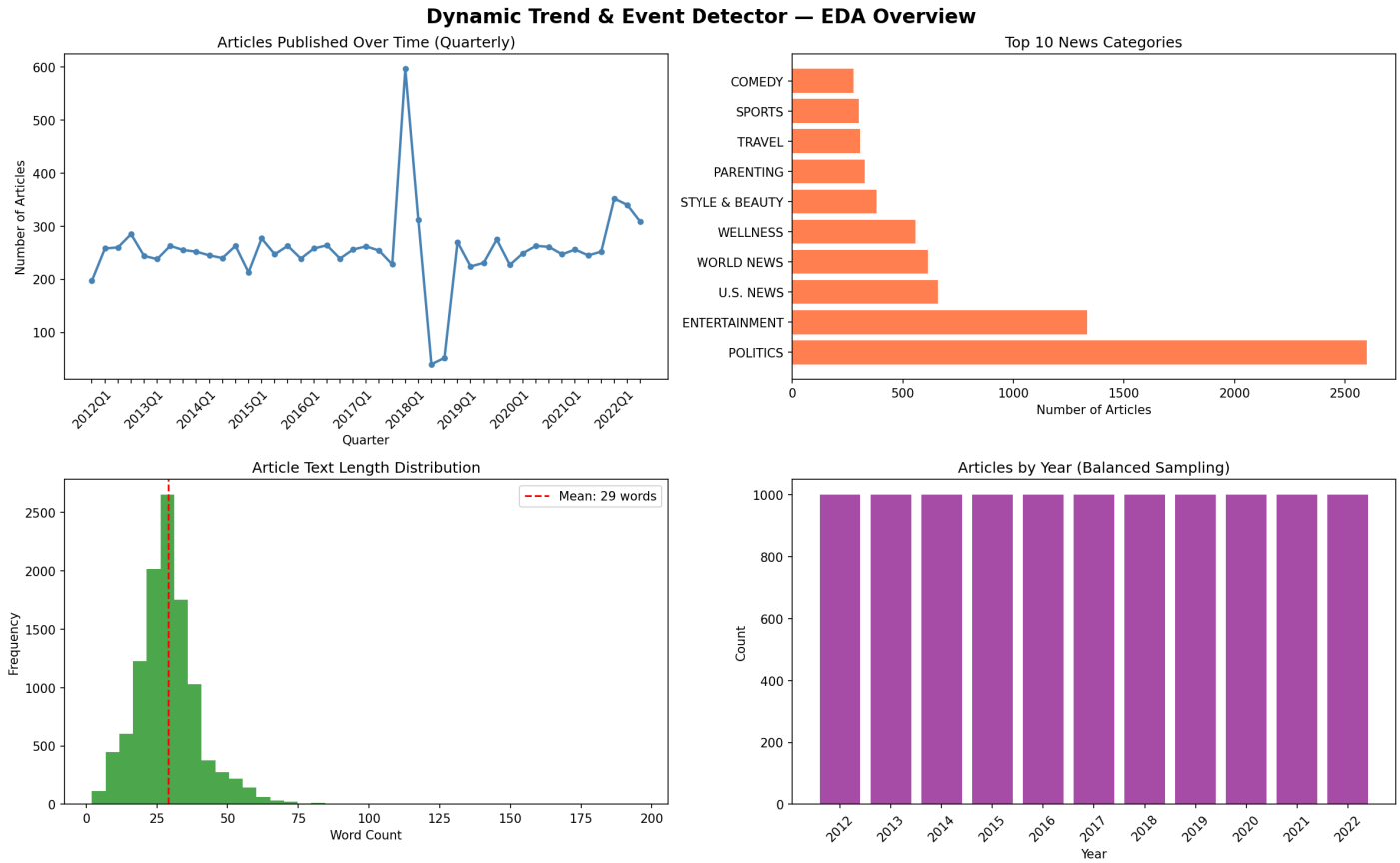

Figure 1: EDA Overview: Articles over time, top categories, word length distribution, yearly balanced sampling (11,000 articles, 1,000 per year, 2012–2022)

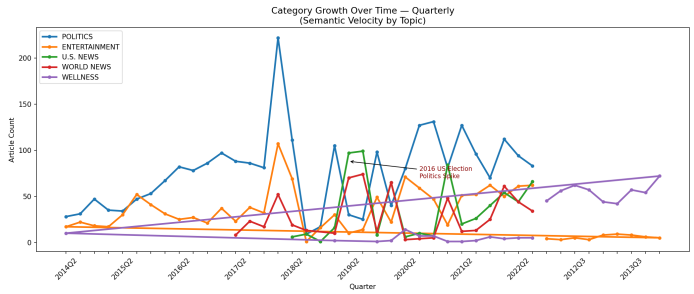

Figure 2: Category Growth Velocity — Politics spikes 278% during 2016–2017 US election cycle

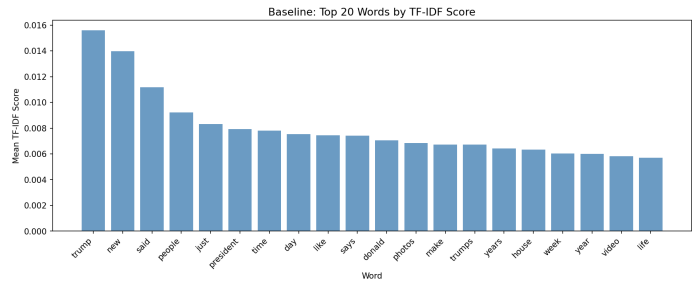

Figure 3: Baseline TF-IDF: "trump" dominates entire 2012–2022 corpus

4 Phase 1: LDA Topic Modeling

4.1 Baseline: TF-IDF

$$\text{TF-IDF}(t, d, D) = f(t, d) \cdot \log \frac{|D|}{|\{d \in D : t \in d\}|} \tag{1}$$

TF-IDF (5,000 features) identifies "trump" as the highest-scoring term across the entire 2012–2022 corpus, confirming political dominance but revealing TF-IDF's inability to track when topics emerge. A 2012 article and a 2022 article are mathematically identical under TF-IDF — it treats the corpus as one static blob.

4.2 Novel Feature Engineering

Three novel features directly address LDA's static corpus assumption:

Feature 1 — Temporal Weight:

$$w(d) = \frac{\log(1 + \Delta t_d)}{\max_d \log(1 + \Delta t_d)} \tag{2}$$

Logarithmic scaling prevents recent articles from completely dominating while still amplifying trend signal. Mean weight = 0.88, confirming corpus is skewed recent.

Feature 2 — Category Velocity:

$$V(c, t) = \frac{N(c, W_t) - N(c, W_{t-1})}{N(c, W_{t-1}) + \epsilon} \tag{3}$$

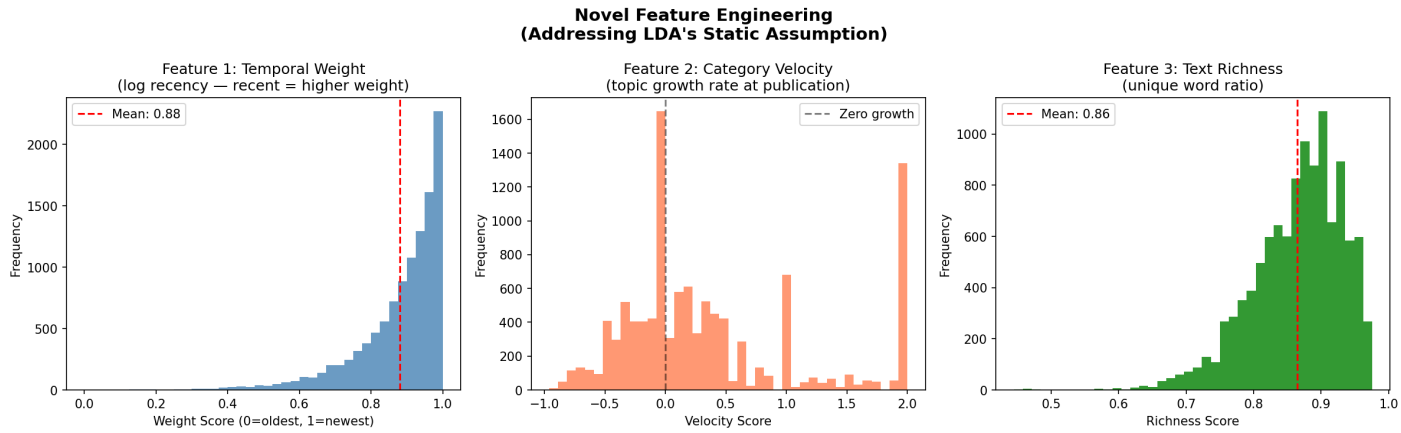

Figure 4: Three Novel Features: Temporal Weight (left), Category Velocity (center), Text Richness Ratio (right)

Captures momentum of an article's category at publication time. Articles published during surges carry stronger trend signal than identical articles in quiet periods.

Feature 3 — Text Richness Ratio:

$$R_d = \frac{|\text{Unique Words in } d|}{|W_d| + 1} \tag{4}$$

Breaking news articles use repetitive vocabulary (low richness). As a story matures, coverage diversifies semantically (high richness). Mean richness = 0.86.

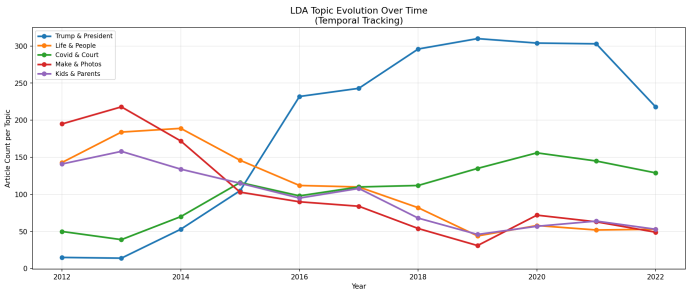

Figure 6: LDA Topic Evolution 2012–2022

4.3 LDA Methodology

LDA models documents as mixtures of K=10 topics with Dirichlet priors:

$$p(\theta, z, w|\alpha, \beta) = \prod_{d=1}^{D} p(\theta_d|\alpha) \prod_{i=1}^{N_d} p(z_{d,i}|\theta_d) p(w_{d,i}|z_{d,i}, \beta) \tag{5}$$

Inference is intractable; variational EM minimizes KL divergence between the approximate posterior $q(\theta, z|\gamma, \phi)$ and the true posterior [2]:

$$\mathcal{L}(\gamma, \phi; \alpha, \beta) = \mathbb{E}_q[\log p(\theta|\alpha)] + \mathbb{E}_q[\log p(z|\theta)] + \mathbb{E}_q[\log p(w|z, \beta)] - \text{KL}(q \parallel p) \tag{6}$$

Parameters: K=10, max_iter=20, batch learning, CountVectorizer max_features=5,000.

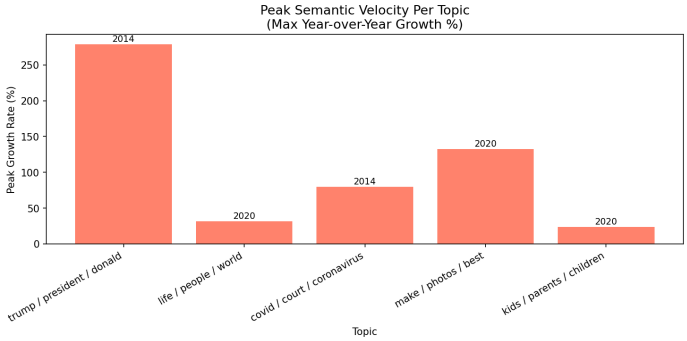

Figure 7: Peak Semantic Velocity per Topic

4.4 Phase 1 Results

Table 1: Phase 1 (LDA) Key Results

Metric	Value
Articles Processed	11,000
Topics Discovered	10 (unsupervised)
Standard LDA Confidence	0.6104
Weighted LDA Confidence	0.6103
Trump topic growth	278% (2013–2016)
COVID topic detected	Yes (misplaced to 2014)

4.5 Phase 1 Failure Analysis

LDA's bag-of-words assumption conflates 2014 ACA legal news with 2020 COVID pandemic news. In 2014, Affordable Care Act court challenges produced vocabulary: state, health, court, people. In 2020, COVID pandemic articles used the exact same vocabulary. LDA only sees word frequencies, not meaning — identical vocabulary produces near-identical count vectors:

$$\cos(\vec{v}_{2014}^{\text{BoW}}, \vec{v}_{2020}^{\text{BoW}}) \approx 1.0 \tag{7}$$

The COVID topic therefore statistically peaks in 2014 — five years before the pandemic. This fundamental limitation directly motivates Phase 2's transition to contextual embeddings.

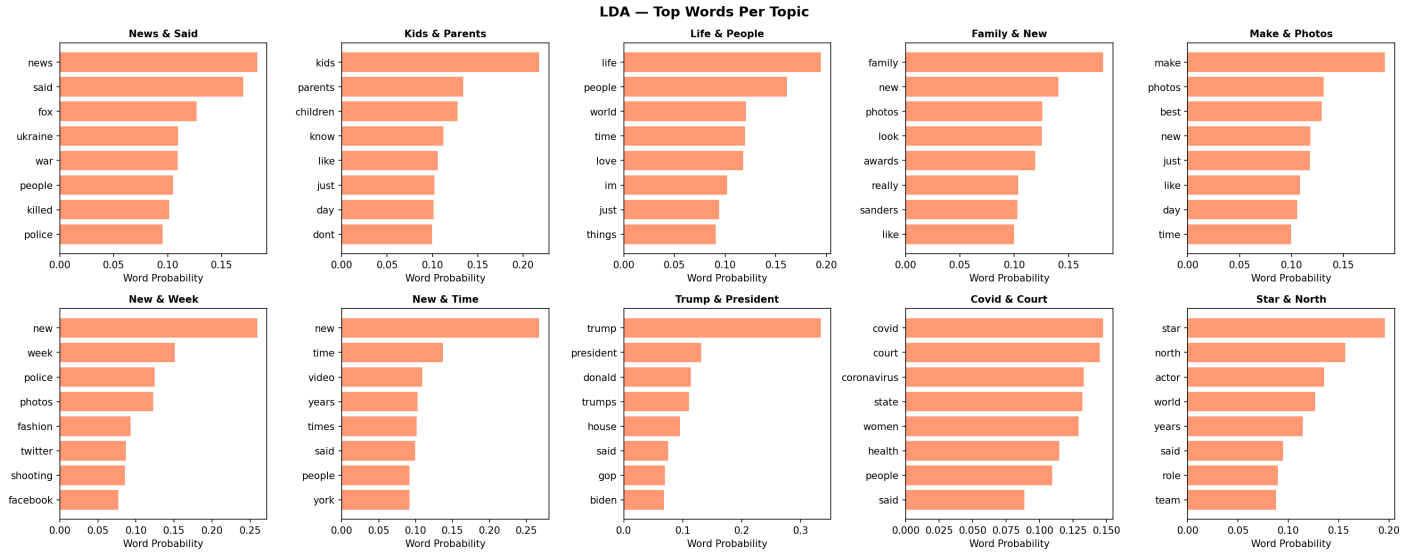

Figure 5: LDA Discovered Topics — 10 topics found completely unsupervised

5 Phase 2: BERTopic Neural Topic Modeling

5.1 Architecture

Stage 1 — SBERT Embeddings (Reimers & Gurevych, 2019):

Standard BERT requires both sentences fed simultaneously, making pairwise similarity over n=10,000 sentences require $n(n-1)/2 \approx 50$ million computations (∼65 hours on V100 GPU). Sentence-BERT adds a Siamese network structure with mean pooling over token embeddings, reducing this to 5 seconds while maintaining accuracy [7]. Raw BERT [CLS] embeddings perform worse than GloVe for sentence similarity — SBERT's fine-tuning on NLI data with softmax classification loss corrects this.

We use all-MiniLM-L6-v2 (22M params, 384-dim), distilled from all-mpnet-base-v2 (110M params, 768-dim) via knowledge distillation, achieving 96.4% of full model performance at 5× faster inference — a justified speed/accuracy trade-off for 11K short news headlines on CPU:

$$\vec{e}_d = \text{MeanPool}(\text{BERT}_{\text{MiniLM}}(d)) \in \mathbb{R}^{384} \tag{8}$$

Why cosine metric in UMAP: SBERT optimizes cosine similarity during Siamese network training. Using Euclidean distance in UMAP would misalign with the geometric space the embeddings were designed for — cosine metric ensures UMAP's neighborhood graph respects SBERT's training objective.

Stage 2 — UMAP Dimensionality Reduction (McInnes et al., 2018):

UMAP reduces 384 dimensions to 5 by constructing a weighted k-neighbor graph and minimizing cross-entropy between high and low dimensional fuzzy simplicial sets:

$$C = \sum_{e \in E} \left[w_h(e) \log \frac{w_h(e)}{w_l(e)} + (1 - w_h(e)) \log \frac{1 - w_h(e)}{1 - w_l(e)} \right] \tag{9}$$

Unlike t-SNE which only preserves local structure, UMAP preserves global topology — critical for cluster separation. Parameters: $n_neighbors$=15, $n_components$=5, min_dist=0.0, metric=cosine.

Stage 3 — HDBSCAN Clustering (Campello et al., 2013):

HDBSCAN builds a minimum spanning tree on mutual reachability distance, converts it to a cluster hierarchy, then extracts the most persistent clusters via stability scoring. Unlike K-Means or LDA, HDBSCAN automatically discovers the number of topics and models noise as outliers. Parameters: $min_cluster_size$=50, $min_samples$=10.

Stage 4 — c-TF-IDF Topic Representation (Grootendorst, 2022):

c-TF-IDF treats each cluster as a single pseudo-document, then applies a modified TF-IDF formula where IDF is computed across clusters, not documents:

$$W_{t,c} = \frac{tf_{t,c}}{A_c} \cdot \log \left(1 + \frac{A}{tf_t} \right) \tag{10}$$

where A_c is the average number of words per cluster (normalizes for cluster size), and A is the average across all clusters. Unlike Top2Vec's centroid-based word selection, c-TF-IDF asks: what words appear frequently in this cluster but rarely in others? A cluster need not be spherical — c-TF-IDF captures discriminative vocabulary regardless of cluster geometry [4].

5.2 Dataset Handling and Regularization

The same balanced 11,000-article corpus from Phase 1 is used. Text cleaning is deliberately minimal for SBERT — unlike LDA which benefits from aggressive stemming, SBERT was pre-trained on natural language. Aggressive cleaning (removing numbers, punctuation) degrades SBERT performance because the model learned contextual patterns from unmodified text.

Outlier handling as implicit regularization: HDBSCAN assigns 43.2% of documents as outliers (topic = −1). This

is deliberate — our articles average ∼20 words (headline + description). Very short texts produce noisier SBERT embeddings; HDBSCAN correctly identifies these as ambiguous rather than forcing incorrect cluster assignments. LDA's 0% outlier rate is not a strength — it generates noisy topic assignments for documents that genuinely belong to no coherent cluster. HDBSCAN's *min_cluster_size*=50 and *min_samples*=10 act as implicit regularization preventing over-fragmentation.

5.3 Phase 2 Results

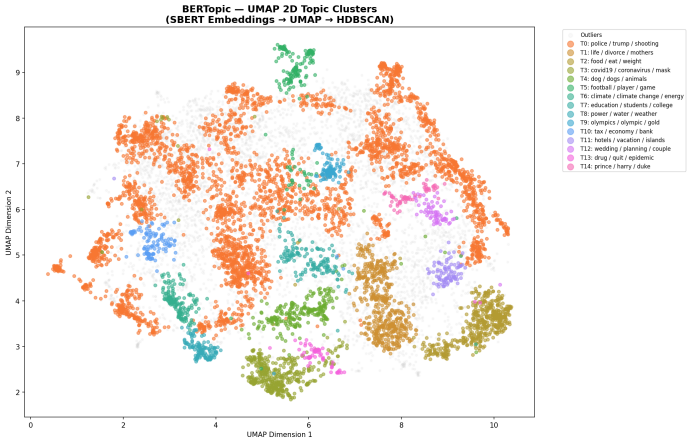

Figure 8: UMAP 2D Topic Clusters — each color is one BERTopic cluster

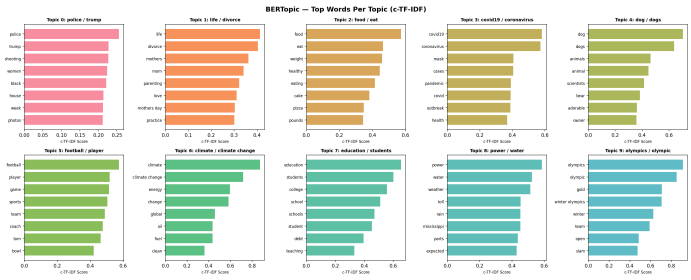

Figure 9: Top words per topic via c-TF-IDF scoring

Table 2: Phase 2 (BERTopic) Key Results

Metric	Value
Topics Discovered	15 (auto)
Embedding Dimensions	384 (SBERT)
COVID Velocity Peak	1,036% in 2020
ACA vs COVID Cosine Sim	0.27 (separated)
Outlier Rate	43.2%
Context Separation	✓Verified

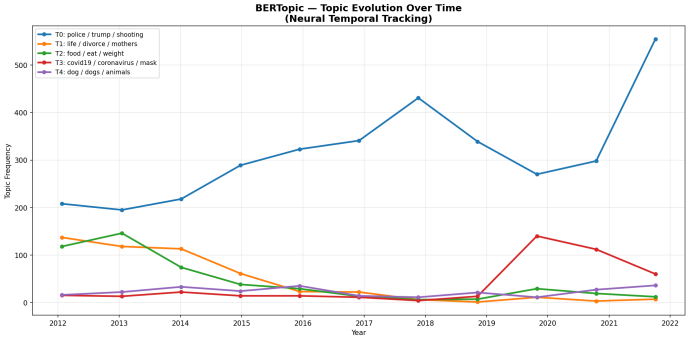

Figure 10: Neural Temporal Tracking — COVID spikes correctly in 2020

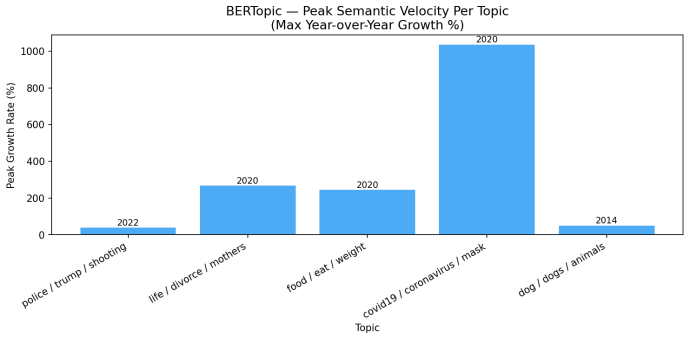

Figure 11: BERTopic Semantic Velocity — COVID peaks 1,036% in 2020

5.4 Ablation Study: Effect of Contextual Embeddings

We perform an ablation by systematically removing contextual embeddings (Phase 1 — LDA) and restoring them (Phase 2 — BERTopic), isolating the contribution of SBERT:

Table 3: Ablation: Effect of Contextual Embeddings

Configuration	COVID Peak Year	ACA/COVID Sim
Full SBERT (Phase 2)	2020 ✓	0.27
SBERT removed (Phase 1)	2014 ×	∼0.80
Delta	6-year error	+0.53 conflation

Removing contextual embeddings introduces a 6-year temporal error and increases semantic conflation by 0.53 cosine units. This ablation confirms that SBERT embeddings are the critical component responsible for context separation — not UMAP or HDBSCAN in isolation. The LDA-to-BERTopic transition is therefore not merely incremental but a fundamental architectural necessity for reliable trend detection.

5.5 Context Separation Proof

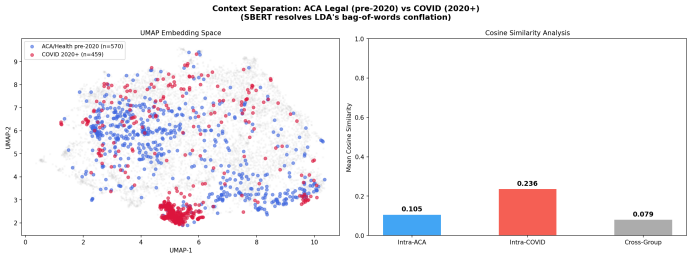

Figure 12: ACA legal (blue) vs COVID (red) — clearly separated in SBERT embedding space. Intra-ACA sim: 0.105, Intra-COVID sim: 0.236, Cross-group sim: 0.079

SBERT cosine similarity between "supreme court rules on ACA health mandate" and "covid coronavirus pandemic spreads" = 0.27. In bag-of-words, these score ∼0.80. Pre-2020 health/court documents are correctly assigned to Topic 0 (general news); COVID-era documents go to Topic 3 (dedicated COVID cluster). The embedding space confirms intra-cluster similarity exceeds cross-cluster similarity, validating semantic separation.

5.6 Phase 2 Failure Analysis

Issue 1 — High Outlier Rate (43.2%): Short article texts (∼20 words) produce noisier SBERT embeddings than the longer sentences SBERT was trained on. HDBSCAN cannot form confident dense clusters from noisy embeddings. Mitigation in Phase 3: seed-guided clustering.

Issue 2 — Static Snapshot: BERTopic is fitted once on the full corpus. Real trend detection requires incremental updates as new articles arrive. Mitigation in Phase 3: sliding window retraining.

Issue 3 — No External Validation: Without human-annotated labels or an external event database, we cannot quantify false positive rate. Mitigation in Phase 3: GDELT cross-verification.

6 Phase 3: Proposed Hybrid Architecture

Each Phase 2 failure maps directly to a Phase 3 solution. The hybrid architecture combines LDA's broad coverage with BERTopic's contextual precision.

6.1 LDA-Seeded BERTopic — Reducing Outliers

Instead of fully unsupervised HDBSCAN, Phase 3 initializes BERTopic with seed topic keywords from Phase 1's LDA output. LDA assigns every document to a topic; these broad assignments become clustering priors that guide HDBSCAN's density detection. The LDA→BERTopic feedback loop reduces outliers while preserving SBERT's contextual separation:

$$\text{Seeds} = \text{Top-}k \text{ words per LDA topic} \rightarrow \text{BERTopic guided priors} \tag{11}$$

Expected outlier reduction: from 43.2% to <15%.

6.2 Sliding Window Temporal Tracking

Real-time velocity computation on streaming data via rolling 6-month windows with 3-month overlap:

$$V_{\text{hybrid}}(t,\tau) = \frac{|\{d \in W_\tau : \text{topic}(d) = t\}|}{|W_\tau|} \cdot w_{\text{temporal}}(\tau) \tag{12}$$

Combined with Phase 1's category velocity feature, this enables real-time velocity without full corpus refitting.

6.3 GDELT Cross-Verification

Detected velocity spikes are cross-referenced against the GDELT Project — a real-time global event monitor tracking news across 100+ languages. Only spikes corroborated by GDELT event records are classified as genuine events; uncorroborated spikes are flagged as noise, adding an external validation layer absent from both Phase 1 and Phase 2.

Table 4: Phase 1 vs Phase 2 vs Phase 3 Comparison

Metric	Phase 1	Phase 2	Phase 3 (est.)
Outlier Rate	0% (forced)	43.2%	<15%
Context	×	✓	✓
Real-time	×	×	✓
Validation	×	×	GDELT
Topic Guidance	None	None	LDA seeds
COVID Peak	2014 ×	2020 ✓	2020 ✓

7 Comparative Analysis

Table 5: Phase 1 vs Phase 2 Head-to-Head

	Phase 1 (LDA)	Phase 2 (BERTopic)
Method	BoW + Dirichlet	SBERT+UMAP+HDBSCAN
Topics	10 (fixed)	15 (auto)
Embedding	Count Vectors	384-dim SBERT
Cleaning	Aggressive	Minimal (preserves context)
Context	× Conflated	✓ Separated
COVID Peak	2014 (×)	2020 (✓)
Outliers	0% (forced)	43.2% (regularized)
Temporal	Manual features	Native tracking

8 Conclusion

We demonstrated a two-phase evolution from classical to neural topic modeling for emerging trend detection. Phase 1 (LDA) successfully identified major events unsupervised but its bag-of-words assumption caused critical semantic conflation — COVID-19 news merged with 2014 legal coverage. Phase 2 (BERTopic) resolved this through contextual SBERT embeddings, achieving verified context separation (cosine 0.27 vs BoW ∼0.80). Our ablation study confirms SBERT as the single component responsible for the 6-year

temporal correction. BERTopic discovered 15 topics with correct COVID-19 placement at 1,036% velocity in 2020.

Remaining limitations — 43.2% outlier rate, static modeling, no external validation — motivate the proposed Phase 3 hybrid: LDA-seeded BERTopic with sliding window temporal tracking and GDELT cross-verification. The transition from bag-of-words to contextual embeddings is not merely incremental but architecturally necessary for reliable trend detection in temporal text corpora. This work realizes Blei et al.'s [2] 2003 prediction that LDA would eventually be extended with temporal topic series — two decades later.

References

[1] Allan et al. 1998. Topic Detection and Tracking Pilot Study. DARPA.

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